

DECODELABS

INDUSTRIAL TRAINING DASHBOARD

Project 1 - Full Stack Development Internship

Professional Project Report

Developer	Mustafa Tahir
Internship Track	Full Stack Development
Organisation	DecodeLabs
Year	2026

HTML5 | CSS3 | JavaScript | Responsive Design | GitHub Pages

1. Project Introduction

The Industrial Training Dashboard is a responsive frontend web application developed for the DecodeLabs Full Stack Development Internship. It brings internship progress, weekly assignments, project milestones and learning development into one structured interface. The final version is presented as a completed Project 1 submission rather than an in-progress prototype.

1.1 Problem and Purpose

Internship activities can easily become scattered across messages, notes and separate task lists. The project addresses this by presenting important training information in a single visual dashboard. It also serves as a practical exercise in semantic HTML, responsive CSS, JavaScript interaction and frontend usability.

1.2 Objectives

- Design a clear and professional internship progress interface.
- Use semantic HTML5 to create meaningful and accessible page structure.
- Use CSS Grid, Flexbox and media queries for responsive layouts.
- Add JavaScript for task filtering, mobile navigation, theme switching and form validation.
- Demonstrate accessibility-aware frontend implementation.
- Prepare clean source code, testing evidence, screenshots and documentation for GitHub submission.

1.3 Final Outcome

The final application shows 100% project completion, six completed development tasks and four completed milestones. It is fully static, requires no backend server and is ready for hosting through GitHub Pages.

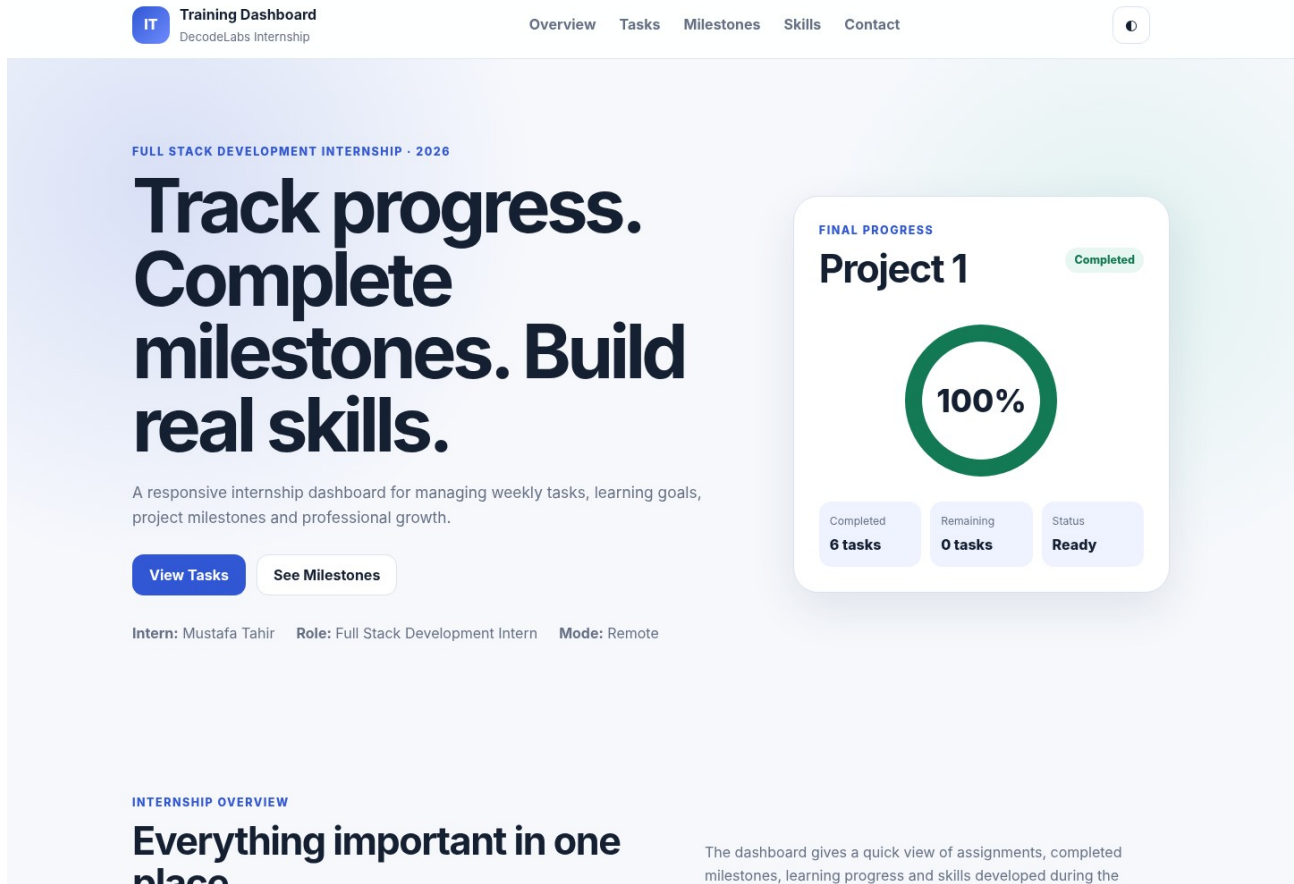
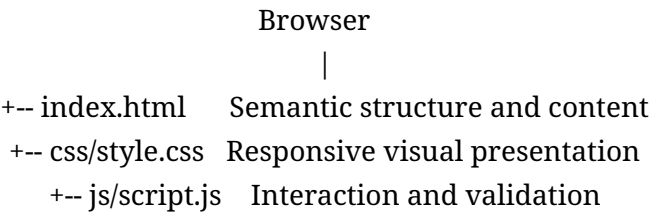


Figure 1. Final dashboard homepage showing Project 1 completion status.

2. Technology Stack and Architecture

Technology	Use in Project	Evidence
HTML5	Semantic structure and page content	Header, navigation, main sections, form and footer
CSS3	Visual design and responsiveness	Grid, Flexbox, media queries, progress components
JavaScript	Client-side interaction	Filters, theme, mobile menu and form validation
Git/GitHub	Version control and repository hosting	Submission-ready folder structure
GitHub Pages	Static deployment target	No server-side runtime required

2.1 Frontend Architecture



2.2 Development Approach

The implementation followed a progressive frontend workflow. Planning and content hierarchy were completed first, followed by semantic HTML structure, responsive styling, JavaScript interactions, accessibility review, testing and final documentation. This sequence is reflected directly in the task board and milestone timeline.

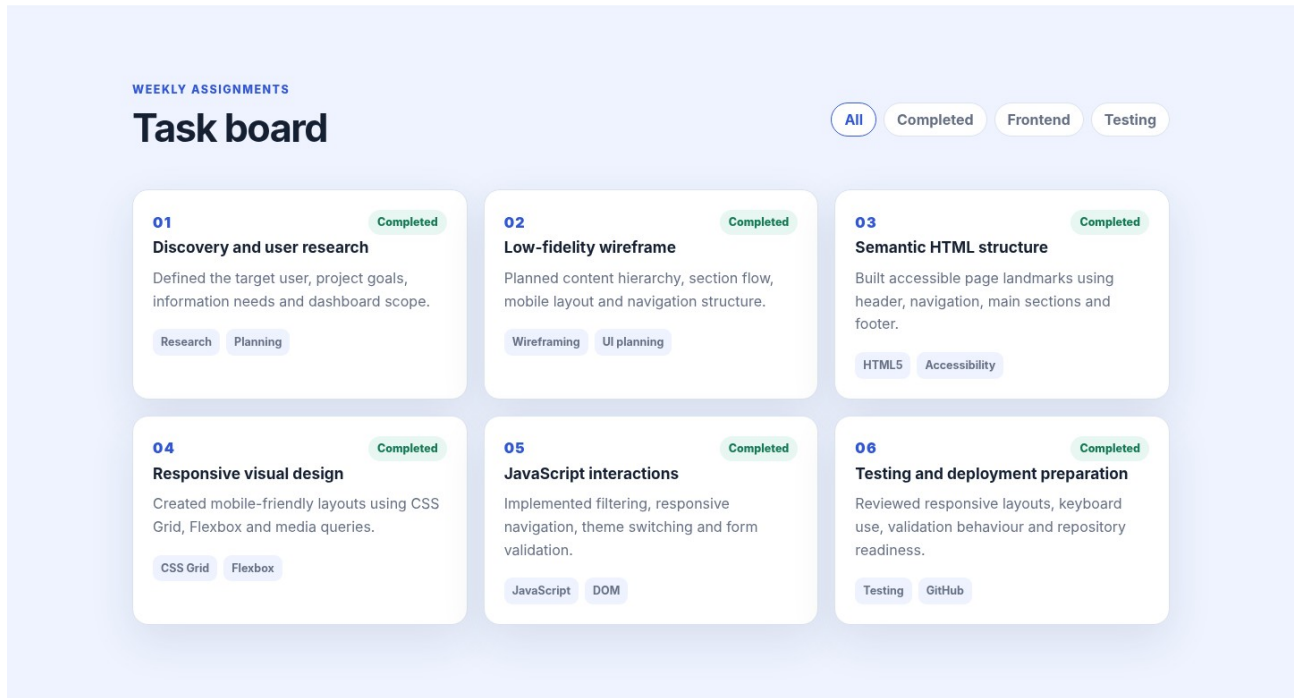


Figure 2. Completed task board documenting the frontend development workflow.

3. Core Features

Completed Progress Summary

The hero section gives an immediate final status with a 100% progress ring, six completed tasks and zero remaining tasks.

Interactive Task Board

Six completed task cards document research, wireframing, semantic HTML, responsive CSS, JavaScript interactions and testing. Filter buttons allow category-based viewing.

Milestone Timeline

A four-stage timeline communicates the project journey from planning through final submission preparation.

Skill Development

Progress indicators communicate development in HTML5, CSS3, JavaScript and accessibility.

Responsive Navigation

Desktop navigation changes into a mobile menu on smaller screens to preserve usability.

Theme Switching

A JavaScript theme control provides light and dark viewing modes and stores the user's preference when browser storage is available.

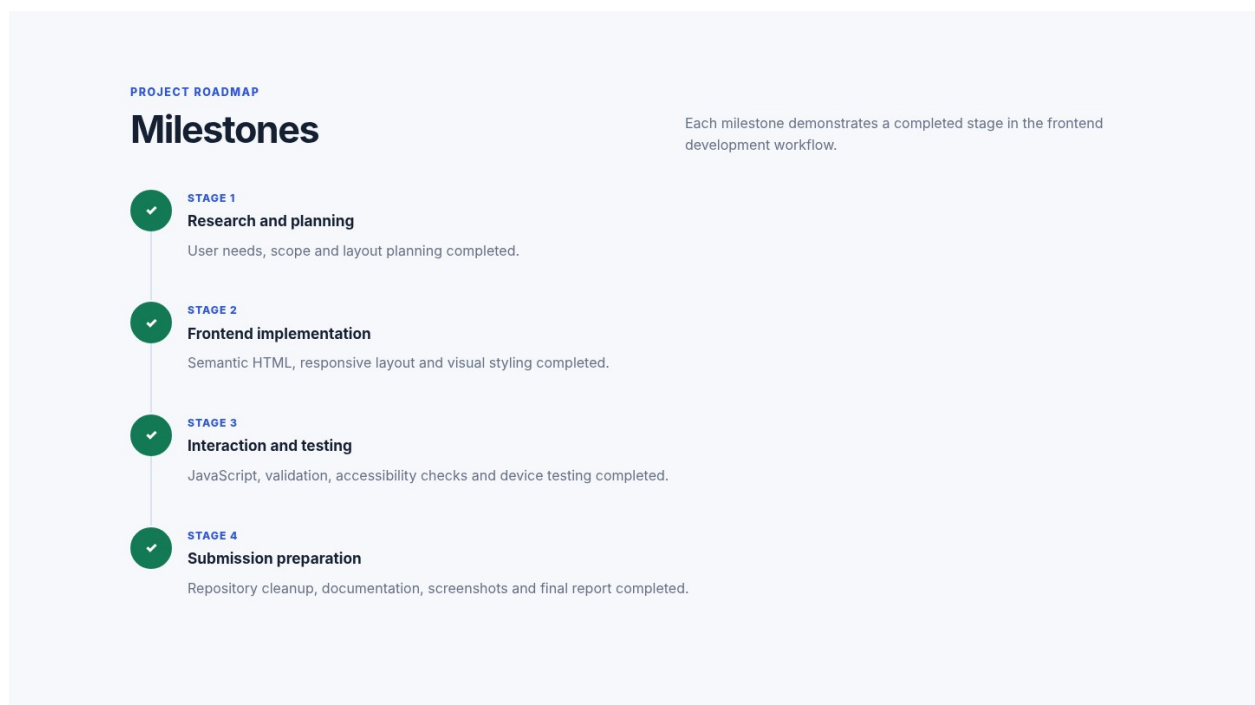


Figure 3. Milestone timeline showing all major development stages as completed.

4. JavaScript, Responsive Design and Accessibility

4.1 JavaScript Functionality

- Task filtering through DOM class changes and data attributes.
- Responsive mobile menu using a button-controlled open state and aria-expanded value.
- Light/dark theme switching.
- Client-side form validation for name, email and project-update length.
- Live error and success feedback.

4.2 Responsive Design

The CSS uses flexible containers, Grid and Flexbox for desktop presentation. At tablet and mobile breakpoints, multi-column layouts collapse and the navigation changes into a compact menu. This allows the same HTML structure to remain usable at different viewport sizes.

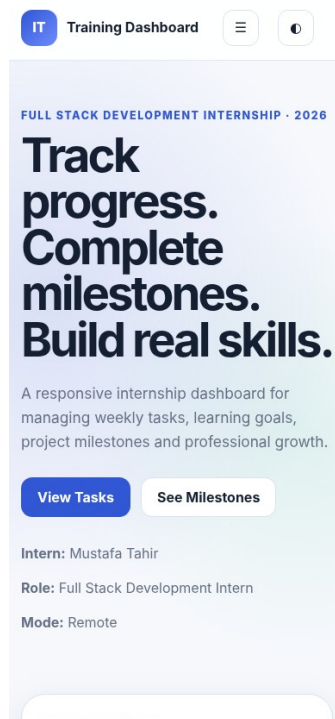


Figure 4. Mobile viewport demonstrating responsive layout behaviour.

4.3 Accessibility Considerations

- Semantic structural elements including header, nav, main, section and footer.
- Skip-to-content link for keyboard users.
- Explicit form labels.
- ARIA labels and aria-expanded state for responsive controls.
- aria-live regions for validation feedback.
- Visible focus styling for form controls and interactive elements.

5. Form Validation and Testing

The project-update form is intentionally frontend-only. It demonstrates input validation but does not transmit or permanently store user data. This limitation is stated within the interface and documentation so the project does not imply backend functionality that is not implemented.

MENTOR COMMUNICATION

Send a project update

This form demonstrates client-side validation. It does not send or store information on a server.

Full name

Please enter at least 2 characters.

Email address

Please enter a valid email address.

Project update

Project update must contain at least 20 characters.

Submit update

Figure 5. Client-side validation feedback when required inputs are missing.

5.1 Testing Summary

Test	Expected Result	Status
Homepage assets	Dashboard opens with styling and JavaScript loaded	PASS
Responsive desktop	Layout remains aligned and readable	PASS
Responsive mobile	Sections collapse correctly	PASS
Task filters	Non-matching task cards are hidden	PASS
Theme switch	Theme state toggles	PASS
Form validation	Invalid inputs show field-level errors	PASS
Valid form	Success feedback appears	PASS
Navigation	Section links and mobile navigation work	PASS

6. Challenges, Solutions and Learning Outcomes

Challenge	Issue	Solution
Responsive layout consistency	Different sections needed to remain balanced on both wide and narrow screens.	Grid and Flexbox layouts were combined with breakpoint-specific rules to create predictable responsive behaviour.
Communicating project status	Earlier versions showed an in-progress state even after implementation was complete.	The final submission version was normalised to show 100% completion, completed tasks and completed milestones.
Form feedback	Browser-only forms can appear functional without clearly communicating validation.	Field-level messages, invalid states and a success status were added while clearly stating that data is not sent to a server.
Accessibility	Interactive controls require more than visual styling.	Labels, ARIA attributes, keyboard-friendly buttons, focus styling and a skip link were incorporated.

6.1 Learning Outcomes

- Improved understanding of semantic HTML and page structure.
- Stronger responsive design skills using Grid, Flexbox and media queries.
- Practical experience with JavaScript DOM manipulation and validation.
- Greater awareness of accessibility requirements in frontend interfaces.
- Experience preparing a project for GitHub, documentation and formal submission.

7. Conclusion and Submission Readiness

Project 1 successfully demonstrates a complete frontend development workflow. The final Industrial Training Dashboard combines responsive layout, semantic structure, JavaScript interaction, validation, accessibility considerations and professional documentation in a single static application.

The packaged submission contains the working source code, screenshots, detailed documentation, testing records, README, editable report and final PDF report. The project can be placed directly inside the DecodeLabs master repository under `Task\_1\_Industrial\_Training\_Dashboard`.

7.1 Submission Package

- Working `index.html`, `css/style.css` and `js/script.js` files.
- Seven evidence screenshots generated from the final project.
- README and detailed project documentation.
- Feature and testing records.
- Submission notes for the DecodeLabs master repository.
- Professional Project 1 report in Word and PDF format.

7.2 Future Improvements

- Connect the update form to a backend API.
- Store tasks and progress dynamically in a database.
- Add authentication and user-specific dashboards.
- Add real internship analytics and chart visualisations.
- Run automated accessibility and cross-browser test suites.

FINAL STATUS: PROJECT 1 COMPLETED AND SUBMISSION-READY

Developer

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